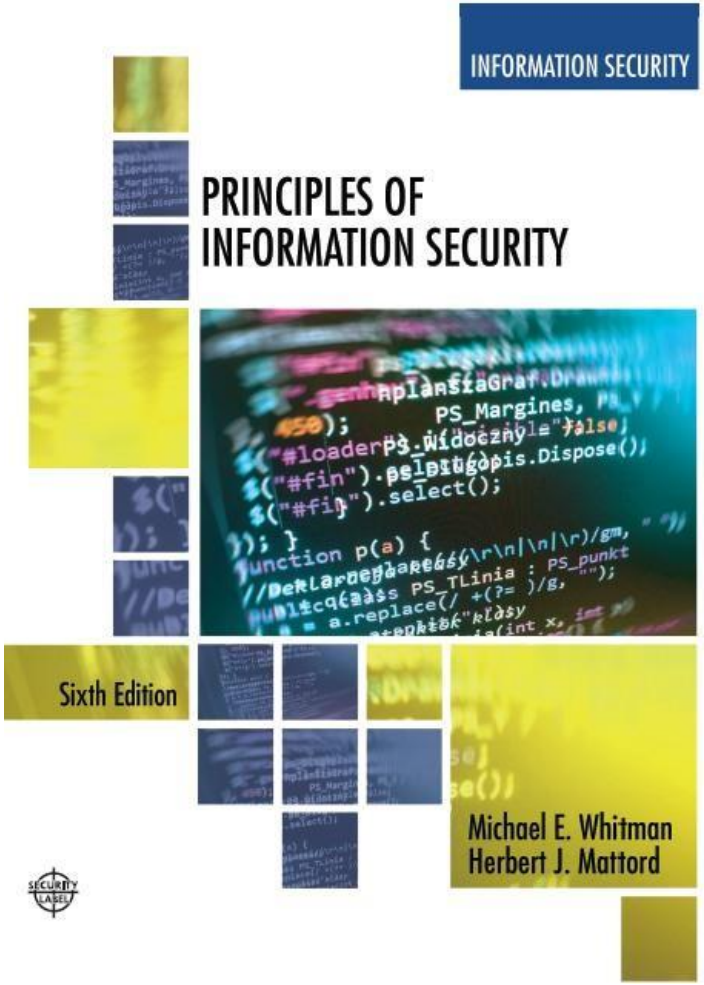
Principles of Information Security

Sixth Edition



Chapter 2

Security Technologies:

- (IDS) and (IPS)
- Cryptography

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Intrusion Detection System (IDS)

- **Intrusion Detection System**(IDS)

  is a network security technology that responsible for monitoring network traffic, analyzing the traffic to identify signatures that match known attacks, and promptly notifying the user in the event of any suspicious activity.

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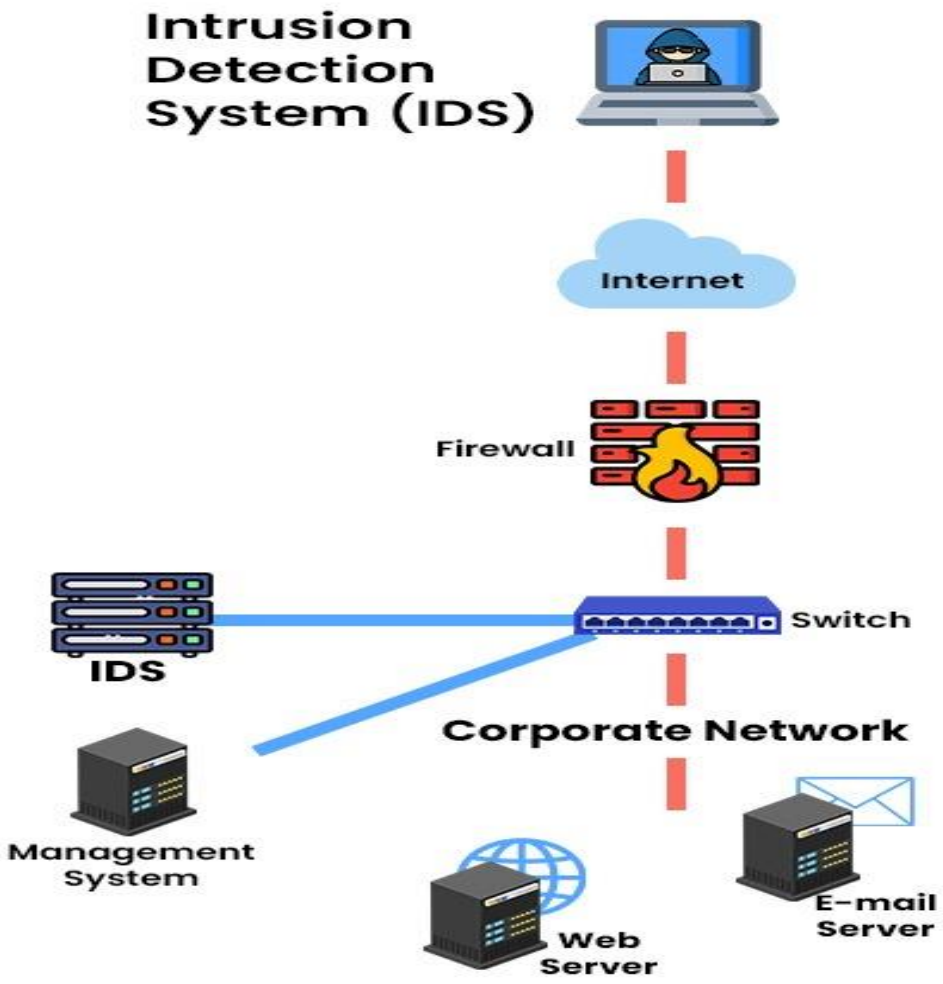
(IDS)



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Types of IDS

1- Network intrusion detection system (NIDS):

It is a kind of IDS that keeps tabs on data packets entering and leaving a network. All traffic or a subset of it might be monitored for risks. Potential threats are evaluated by an NIDS against patterns of hazardous activity. Due to its superior coverage compared to host-based systems, this is the favoured choice for large organizations.

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Types of IDS

2- Host Intrusion Detection System (HIDS):

A HIDS is a kind of IDS that is installed on and only watches over one host at a time (such as a computer). In addition to traffic, client behavior on the computer is also tracked. HIDS compares the host's current checksum to a previously calculated one and notifies the system administrator if any difference is found.

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Types of IDS

Note:

It is important to note that both NIDS and HIDS can work side by side. With HIDS and NIDS working together, sensitive devices or workstations will have further protection from potential threats. While malicious software may be able to bypass a NIDS, their activity will be detected by a HIDS.

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Intrusion Prevention System (IPS)

- An intrusion prevention system (IPS) is a network security technology that responsible for the monitoring of network traffic as well as blocking the infected packets or the server from where the packet is transmitting.

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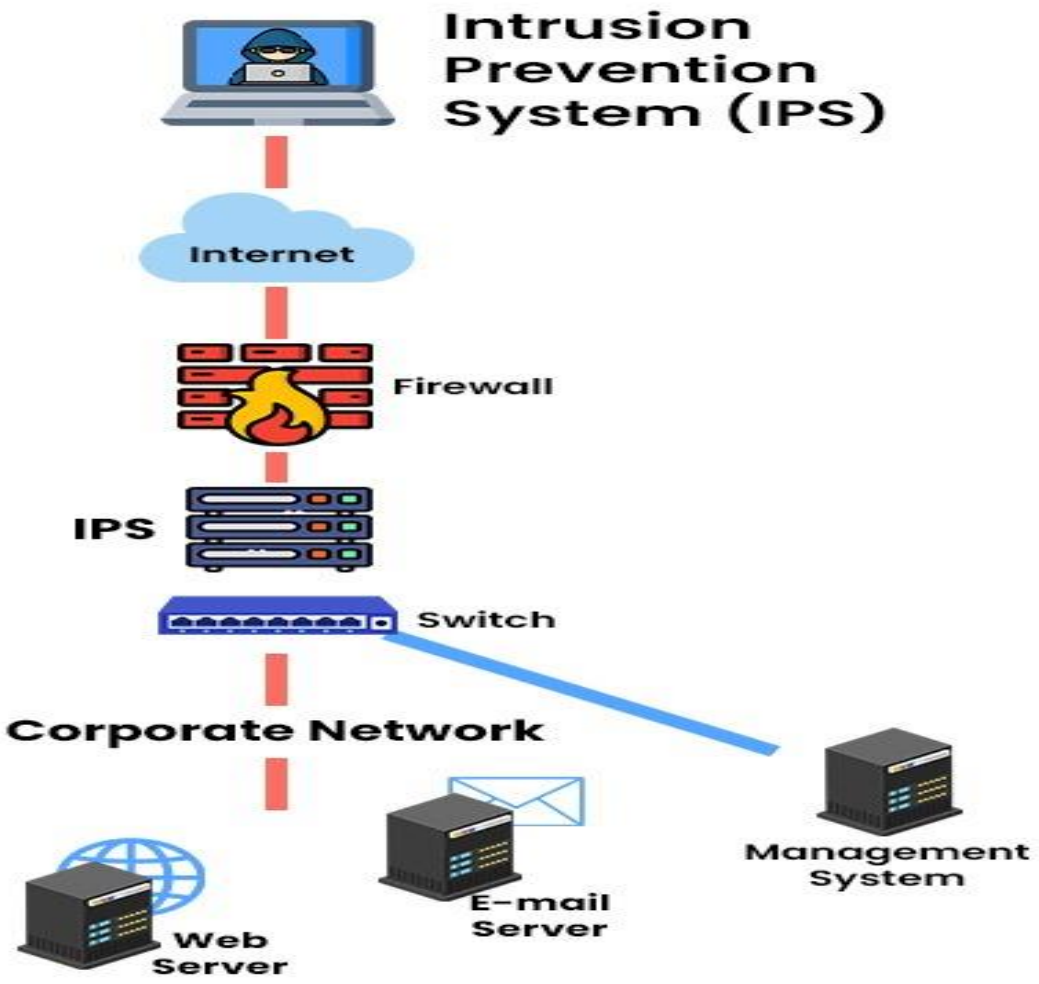
(IPS)



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Types of IPS

1. **Network-based Intrusion prevention system (NIPS):**

   It is deployed at strategic points in the network, such as routers or firewalls, and analyzes all incoming and outgoing traffic.

2. **Host-based intrusion prevention system (HIPS):**

   It is installed on individual hosts, such as servers or workstations, and monitors the system's behavior and processes.

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Types of IPS.

3. Wireless intrusion prevention system (WIPS):

It is designed to protect wireless networks from unauthorized access and attacks. In general, WIPS has two components, namely overlay monitoring and integrated monitoring.

4. Network behavior analysis (NBA):

It is a type of IPS that uses statistical analysis and machine learning to identify abnormal network patterns and anomalies. In simple words, It is similar to NIPS, but it works on anomalies.

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Difference between IDS and IPS

Factors	Intrusion Detection System (IDS)	Intrusion Prevention System (IPS)
Function	IDS only alerts the network administrator when it detects an intrusion.	IPS actively blocks or drops the malicious packets before they reach the target.
Placement	IDS is usually placed outside the network perimeter, such as behind a firewall or a router.	IPS is usually placed inside the network perimeter, such as between a firewall and a switch.
System Type	Passive as it only monitors and then notifies the administrator.	Active as it monitor as well automatically defends the network.
Anomaly Response	Sends a notification to the user or log	Drops or modifies malicious packets
Performance	Low impact on the network speed as it only detects the intrusion.	High impact on network speed as it has to analyze and modify or block traffic in real time.

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Note:

Firewalls assist in blocking as well as filtering network traffic, whereas IDS is only for the detection of intrusions, and IPS assists in blocking as well as alerting of intrusions.

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Cryptography

Cryptography

Is the science of concealing information so that no one else can see it except the intended recipient. The cryptographic practice entails using an encryption algorithm to convert plaintext to ciphertext. The receiver deciphers the ciphertext text using a shared or determined key.

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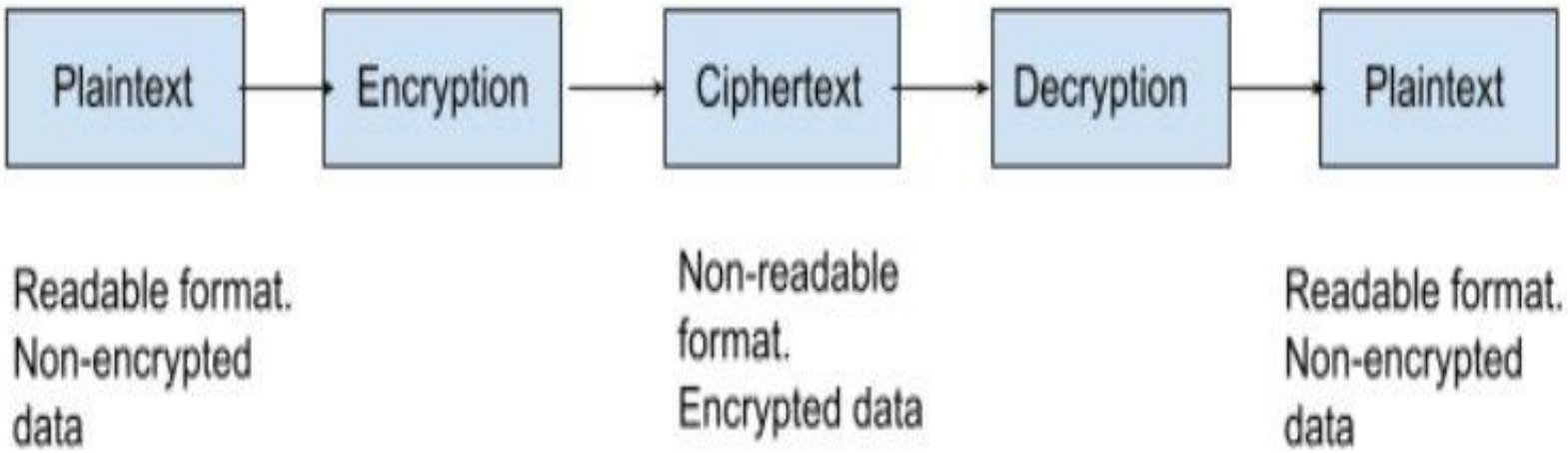
Cryptography



Cryptography

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Types of Cryptography

1- Symmetric Key Cryptography

Symmetric key cryptography is also known as secret-key cryptography, and in this type of cryptography, you can use only a single key. The sender and the receiver can use that single key to encrypt and decrypt a message. Because there is only one key for encryption and decryption, the symmetric key system has one major disadvantage: the two parties must exchange the key in a secure manner. An example of symmetric key cryptography is Blowfish.

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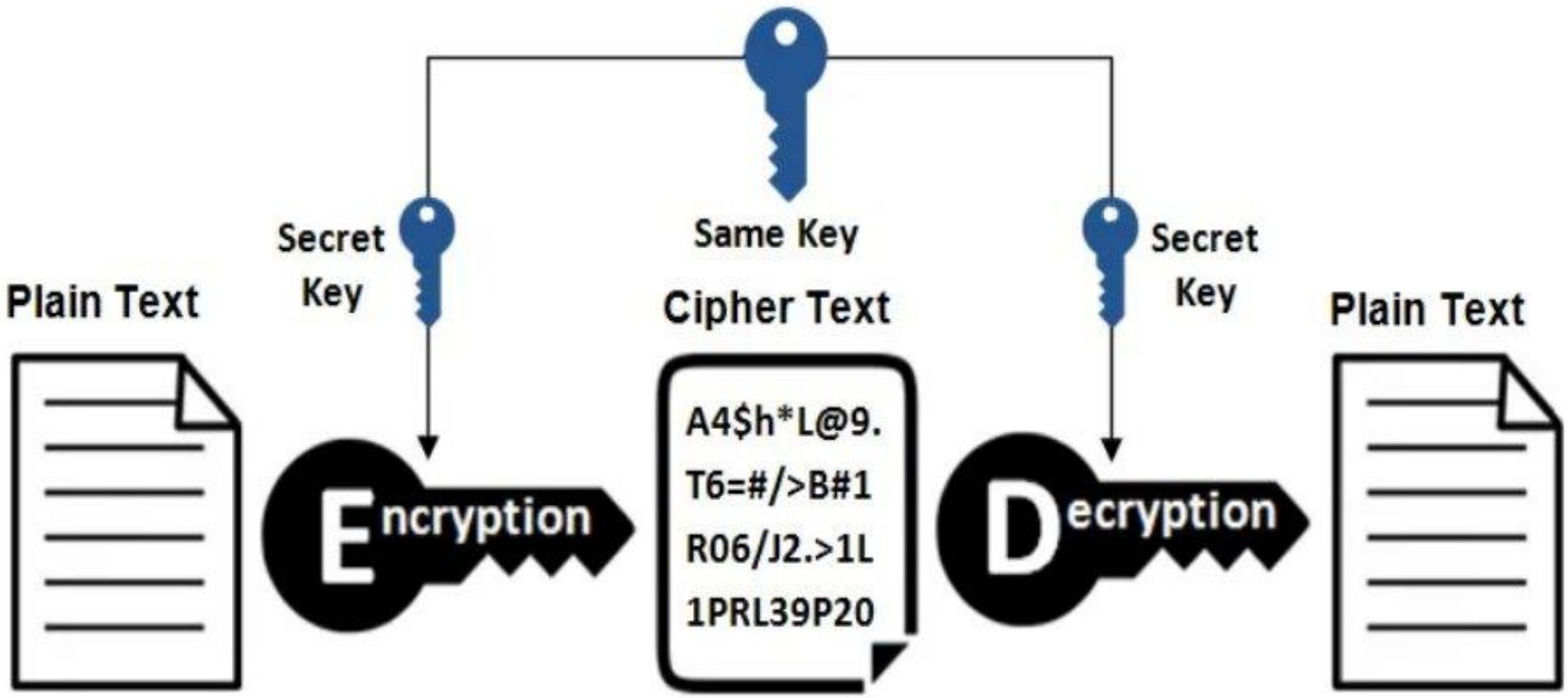
Types of Cryptography



Symmetric key cryptography

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Types of Cryptography

2- Asymmetric Key Cryptography

Asymmetric key cryptography is also known as public-key cryptography, and it employs the use of two keys. This cryptography differs from and is more secure than symmetric key cryptography. In this system, each user encrypts and decrypts using two keys or a pair of keys (private key and public key). Each user keeps the private key secret and the public key is distributed across the network so that anyone can use those public keys to send a message to any other user. You can use any of those keys to encrypt the message and can use the remaining key for decryption. An RSA algorithm is an example of asymmetric key cryptography.

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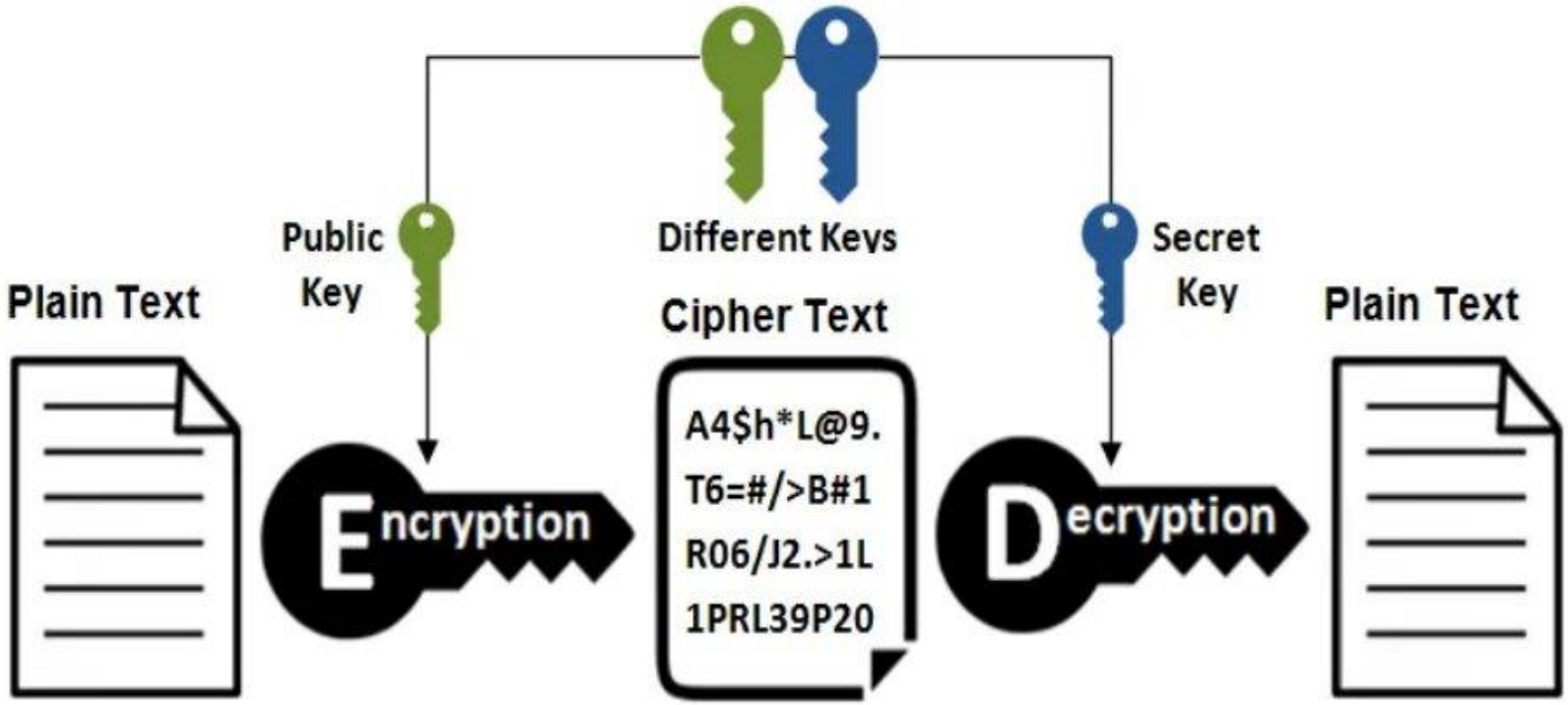
Types of Cryptography



Asymmetric key cryptography

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Types of Cryptography

3- Hash Function

This algorithm makes no use of any keys. A hash value with a fixed length is calculated based on the plain text, making it impossible to recover the plain text's contents. Many operating systems encrypt passwords using hash functions.

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Types of Cryptography Algorithm

- Cryptographic algorithms are primarily of two types, and you can use them for critical tasks, such as authentication, data encryption, and digital signatures.

1- RSA: RSA is an asymmetric cryptographic algorithm based on the block cipher principle. It converts plain text to ciphertext at the receiver end and vice versa. If we use User A's public key for encryption, we must use the same user's private key for decryption.

2- DES: Data Encryption Standard (DES) is a symmetric cipher algorithm that encrypts and decrypts data using the block cipher method. The algorithm uses 48-bit keys to convert the plain text in 64-bit blocks into ciphertext. It operates on the Fiesta Cipher Structure.

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How Does Cryptography Work?

Cryptographic algorithms are central to how cryptography works. Cryptographic algorithms, also called ciphers, are mathematical functions that encrypt text by combining them with keys such as phrases, digits, words, and so on. You can define the effectiveness by the strength of the cryptographic algorithms and the level of key secrecy.

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Applications of Cryptography

There are various applications of cryptography. Some of those applications are:

- **Confidentiality:** Cryptography allows users to store encrypted data, avoiding the major flaw of hacker circumvention.
- **Non-repudiation:** The creator/sender of information cannot later deny his intent to send information.
- **Authentication:** Helps to authenticate the sender and receiver's identities along with the destination and origin of the information.
- **Integrity:** Information cannot be altered during storage or in transit between the sender and the intended receiver without any addition to the information being detected.

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